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Applied Energy

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About the journal

Aims and scope

Applied Energy provides a forum for information on research, innovation, development, and demonstration in the areas of **energy conversion** and **conservation**, the optimal use of **energy resources,** analysis and optimization of **energy processes**, **multi-energy systems**, mitigation of **environmental pollutants through sustainable, secure, efficient energy systems, and fair and efficient transition to a Net Zero future**. The journal publishes original research papers, review articles, technology assessment, bibliographical studies, and perspective papers. Authors are encouraged to submit manuscripts which bridge the gaps between research, development, and implementation. The breadth of coverage ranges from innovative technologies and systems of both low-carbon and renewable energy to the economic industrial, commercial, and domestic use of energy with no or minor impact on the environment. *Applied Energy* is also concerned with the energy related modelling, forecasting, and decision-making problems, energy market design and operation, energy conservation strategies, power system planning and operation, energy users modelling, and the environmental, social and economic impacts of energy policies and usage, including climate change mitigation and other environmental pollution reduction.

Key topics covered in the journal include, but not limited to (the focus must be on applied aspects of energy):

- Renewable energy resources and technologies
- Energy networks and Smart Grid
- Microgrids and local energy systems
- Energy systems for Carbon Neutrality (climate mitigation and adaptation)
- Integrated multi-energy systems and sector coupling
- Energy in buildings
- Energy storage
- Hydrogen and sustainable fuels
- Energy conservation and energy efficiency
- Flexibility, demand response, and forecasting models

Which articles connect with your research?

Describe your research question and surface relevant articles from this journal and related peer-reviewed literature.

- Carbon capture, utilization and storage for energy systems
- Applications of Artificial Intelligence in energy
- Optimization and decision-making methods for energy systems problems
- Data analytics for energy problems
- Decarbonization of transport (including EVs and hydrogen vehicles)
- Decarbonization of heat
- Process systems and decarbonization of industry
- Energy materials and emerging energy technologies
- Urban energy and smart city
- Cyberphysical systems for energy applications
- Energy markets and economics
- Energy communities
- Virtual Power Plants and other energy/power related aggregators
- Energy policy and societal challenges

Applied Energy is the companion title to the open access journal Advances in Applied Energy

This journal welcomes contributions that support and advance the UN's sustainable development goals, in particular SDG 7 (Affordable and clean energy)

Article types

Full Length Research Article: Full length articles are original, high-quality research papers presenting novel scientific findings.

Review Articles: A review article provides a comprehensive and critical analysis of existing research and literature on a particular topic. It aims to summarize, synthesize, and evaluate the current state of knowledge in a specific field. The primary goal is to offer readers a thorough understanding of the subject area, identify research gaps, and suggest potential directions for future studies. Applied Energy invites subject reviews from accomplished researchers and scholars in a broad range of topics within the scope of the journal. Reviews are to be in-depth, critical analyses from authors who have the demonstrated track record and can write with authority on the proposed topic. Reviews may be undertaken on a broad subject area or on a very specific topic and are expected to be more than just a survey of the literature accompanied by a long list of references. Applied Energy welcomes reviews that present the state-of-the-art and the critical issues that have been solved and those challenges that remain unsolved. We particularly appreciate reviews that provide insights about where the current research is heading and those issues that attract significant research and development in the near and far future.

The Review articles will be evaluated based on the following criteria:

(i) The proposed review topic aligns with the scope of Applied Energy.
(ii) The author team should have a demonstrated track record and can write with auth
articles authored by a multi-nation and/or multi-disciplinary team.
(iii) There are critical reviews on research advancements related to the topic, identific
particular domain, critical assessment of breakthrough innovations and applications
landmark implications, and strategic energy-related research direction on a global sc
(iv) The specific review gaps in the review literature which need to be addressed have
(v) The scientific, technical and policy issues and challenges have been clearly covered in the proposed review.
(vi) The outlook of the proposed review topic is clearly discussed in terms of its anticipated research trajectory, significance, and impacts.

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(vii) Any other relevant comments that highlight the salient contributions to be made by the proposed review.

Technology assessment articles: A technology assessment paper evaluates the impact, consequences, and implications of specific technologies on society, the environment, and various aspects of human life. This type of paper goes beyond a mere review of existing literature and delves into the broader societal, ethical, economic, and policy dimensions of the technology in question. Technology assessment papers analyse the benefits, risks, challenges, and potential ramifications of the technology, aiming to provide evidence-based guidance for decision-makers and stakeholders.

Perspective papers: A perspective paper, also known as an opinion paper, position paper, or viewpoint paper, presents the personal viewpoint or opinion of the author(s) on a specific topic. Unlike a review paper, which provides an objective analysis of existing literature, a perspective paper allows authors to express their insights, interpretations, and viewpoints regarding a particular issue or concept. It is more subjective in nature and may include personal reflections, criticisms, or proposed conceptual frameworks. Perspective papers often stimulate discussions and debate within the academic community, providing alternative perspectives on significant topics.

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- The article's publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out.
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1. The conception and design of the study, or acquisition of data, or analysis and interpretation of data.
2. Drafting the article or revising it critically for important intellectual content.
3. Final approval of the version to be submitted.

All authors are accountable for all aspects of the work, to ensure that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Inclusive language acknowledges diversity, conveys respect to all people, is sensitive to differences, and promotes equal opportunities. Authors should ensure their work uses inclusive language throughout and contains nothing that might imply one individual is superior to another on the grounds of:

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- gender
- race
- ethnicity
- culture
- sexual orientation
- disability or health condition

Avoid the use of descriptors about personal attributes unless they are relevant and valid. Write for gender neutrality with the use of plural nouns ("clinicians, patients/clients") as default. Wherever possible, avoid using "he, she," or "he/she."

Don't make assumptions about the beliefs of readers and keep writing free from bias, stereotypes, slang, reference to dominant culture, and cultural assumptions.

These guidelines are meant as a point of reference to help you identify appropriate language but are by no means exhaustive or definitive.

Reporting sex- and gender-based analyses

There is no single, universally agreed-upon set of guidelines for defining sex and gender. We offer the following guidance:

- Sex- and gender-based analyses (SGBA) should be integrated into research design when research involves or pertains to humans, animals, or eukaryotic cells. This should be done in accordance with any requirem within a field.
- Sex and/or gender dimensions of the research should be addressed within the arti the research.
- Explicitly state applied definitions of sex and/or gender. This enhances precision, r confusing terms and the constructs they refer to.

We advise you to read the Sex and Gender Equity in Research (SAGER) guidelines and t offer systematic approaches to the use of sex and gender information in study design interpretation.

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For further information we suggest reading the rationale behind and recommended use of the SAGER guidelines.

Definitions of sex and/or gender

We ask authors to define how sex and gender have been used in their research and publication. Some guidance:

- Sex generally refers to a set of biological attributes that are associated with physical and physiological features such as chromosomal genotype, hormonal levels, and internal and external anatomy. A binary sex categorization (male/female) is usually designated at birth ("sex assigned at birth") and is in most cases based solely on the visible external anatomy of a newborn. In reality, sex categorizations include people who are intersex/have differences of sex development (DSD).

- Gender generally refers to socially constructed roles, behaviors, and identities of women, men, and gender-diverse people that occur in a historical and cultural context and may vary across societies and over time. Gender influences how people view themselves and each other, how they behave and interact, and how power is distributed in society.

Depending on the focus of a paper, sex and/or gender may or may not be relevant to the content of the paper. We recognize that beliefs, attitudes, and laws relating to sex and gender may vary. These articles do not attempt to dictate author beliefs but rather require that, where relevant to an author's research or paper, the author must provide clear explanations of how the paper and research define and use sex and gender.

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institutional name so that the institutional name can be independently verified fo

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Authors must follow ethical guidelines for research involving humans, their data, and
Authors should ensure that they adhere to the following:

- All studies must be performed in compliance with relevant laws, regulations, and must be approved by the appropriate institutional committee(s). For instance, medical research involving humans should be carried out in accordance with the World Medical Association Declaration of Helsinki: Ethical principles for medical research involving human participants and should follow the International Committee of Medical Journal Editors (ICMJE) recommendations for the conduct, reporting, editing and publication of scholarly work in medical journals and aim to be representative of human populations in terms of sex, age, and ethnicity. Sex and gender terms should be used correctly, as outlined in the "Reporting sex- and gender-based analyses" section of this guide for authors.

- Informed consent must be obtained for research with human participants, including their data or biological materials. Consent must be given without coercion, and the privacy rights of human participants must always be observed. If your study contains human participants, please read Elsevier's full human research policy for further information.

    - Authors should be aware that written (or electronic) consent is required for certain studies under national laws. In studies where only verbal informed consent was obtained, authors should include justification for the absence of written consent, the name of the approving ethics committee, and a statement on how verbal consent was recorded.

    - Consent for publication should also be obtained.

- Authors must obtain and retain explicit written consent for any identifiable participant materials (e.g., case details, videos, recordings, images, photographs, and illustrations). All participant information must be de-identified: do not include names or personal identifiers (e.g., initials, ID numbers, DOB)

- Images should be used only when scientifically necessary and with explicit consent, and nonessential identifying details must be removed. If identifiers are altered to ensure anonymity, authors must confirm that scientific meaning is not affected. Fully anonymized images (e.g., x-rays, scans) that contain no identifying marks and are not accompanied by identifying text do not require formal consent, though informing the patient is good practice. Case reports always require consent. Simply obscuring identity (e.g., eye bars or blurring) is not sufficient without consent.

- Written consents must be retained by the authors. They should not be provided to this journal unless this is specifically requested in exceptional circumstances, for example, when a legal issue arises. Only then should you provide copies of the consents, or evidence that all relevant consents were obtained.

- Appropriate consents and safeguards should be in place for research with vulnerable populations, as described in Elsevier's human research policy. In many countries, additional ethical approval may be needed for research with those who lack the capacity to consent (e.g., children or those with mental health or neurological conditions including dementia).

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All studies involving human participants require a statement that details any ethical approval(s), guidelines followed, confirmation of informed consent, and any other necessary information in line with Elsevier's human research policy. The statement should contain the date and reference number of any ethical approval(s) obtained.

If ethical approval is not required by national laws or where a study has been granted an exemption from an ethics committee, the statement must explain why the study is exempt and confirm that the appropriate protocols for protecting the rights and privacy of all participants were utilized during the execution of the research (e.g., no coercion to participate, full disclosure of study requirements and risks, consent of participants (or their legal representatives), no release of participant data without their knowledge, etc.).

Human biological material

For all studies that use human organs or tissues, sufficient evidence must be provided
Principles on Human Cell, Tissue and Organ Transplantation. The source of the organs
transparent and traceable and must not be sourced from executed prisoners or prison
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For clinical studies, a statement that informed consent was obtained from the patient
approval from a suitable institution, may be sufficient evidence. The journal has the ri
feels this is needed.

If the manuscript describes organ transplantation, authors must additionally declare

- Autonomous consent, free from coercion, was obtained from the donor(s) or their next of kin.

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- Organs and/or tissues were not sourced from executed prisoners or prisoners of conscience.

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You are required to provide a concise and factual abstract that does not exceed 250 w

- State the purpose of your paper.
- Outline basic procedures followed.
- Include your main findings, their statistical significance, where applicable, and your principal conclusions.
- Emphasize new and important aspects of your study or observations.

Some guidelines:

- Abstracts must be able to stand alone, as abstracts are often presented separately from the article.
- Avoid references. If any are essential to include, ensure that you cite the author(s) and year(s).
- Avoid non-standard or uncommon abbreviations. If any are essential to include, ensure they are defined within your abstract at first mention.

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You are required to provide 1 to 7 keywords for indexing purposes. Keywords should be written in English. Try to avoid keywords consisting of multiple words (using "and" or "of").

Only use abbreviations in keywords if they are firmly established in the field.

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You are required to provide article highlights at submission.

Highlights are a short collection of bullet points that should capture the novel results of your research as well as any new methods used during your study. Highlights will help increase the discoverability of your article via search engines. Some guidelines:

- Submit highlights as a separate editable file in the online submission system with the word "highlights" included in the file name.
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You are encouraged to provide a graphical abstract at submission.

The graphical abstract summarizes the contents of your article in a concise, pictorial, professional form, highlighting important aspects of the research. It is designed to appeal to an interdisciplinary audience and draw more attention to your online article. Some guidelines:

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- Preferred file types for graphical abstracts are TIFF, EPS, PDF, or MS Office files.

More information, examples, and a template can be found in the graphical abstract g

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- Submit math equations as editable text, not as images.
- Present simple formulae in line with normal text, where possible.
- Use the solidus (/) instead of a horizontal line for small fractional terms such as X/Y.

- Present variables in italics.
- Denote powers of e by exp.
- Display equations separately from your text, numbering them consecutively in the order they are referred to within your text.

Tables

Tables must be submitted as editable text, not as images. Some guidelines:

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- Number tables consecutively according to their appearance in the text.
- Please provide captions along with the tables.
- Place any table notes below the table body.
- Avoid vertical rules and shading within table cells.

Use tables sparingly, ensuring that any data presented in tables is not duplicating results described elsewhere in the article.

Figures, images, and other artwork

Figures, images, diagrams, and other graphical media (artwork) must be supplied as separate files along with the manuscript, in line with these artwork and media instructions. Some excerpts:

When submitting artwork:

- Cite all images in the manuscript text.
- Number images according to the order they appear within your article.
- Submit each image as a separate file using a logical naming convention for your files (for example, Figure_1, Figure_2 etc.).
- Text graphics may be embedded in the text at the appropriate position. If you are working with LaTeX, text graphics may also be embedded in the file.

Captions

All artwork must have a caption. A caption should consist of a brief title (not displayed on the figure itself) and a description of the image. Keep the amount of text to a minimum, though any symbol or abbreviation should be explained.

More information about adding captions to your submission..

Formats

Submit your artwork in the formats listed below, following the given requirements:

Image type	Recommended file type
Vector drawings	EPS, PDF
Photographs (color or grayscale)	TIFF, JPG, PNG

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Describe your research question and surface relevant articles from this journal and related peer-reviewed literature.

Bitmapped line drawings	TIFF, JPG, PNG	Minimum resolution 1000 dpi (min. width: single column: 3543 pixels; full page: 7480 pixels)
Charts and large or shaded tables	XLS, XLSX	Supply in a clearly readable format
Annotated images	PPT, PPTX	Annotations and text must be legible; Minimum resolution of embedded images 300 dpi

Do not submit:

- Files that are too low in resolution.
- Artwork where text is disproportionately small compared to the image size, as text may become unreadable.
- Multiple images or graphs combined into one file, as this affects legibility and accessibility. An exception may be made for images that are intended to be compared to each other, such as before, during, and after photographs.

Color artwork

If you submit usable color figures with your accepted article, we will ensure that they appear in color online. Please ensure that color images are accessible to all, including those with impaired color vision. Learn more about color and web accessibility.

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- AI tools may be used to support the creation of explanatory images (such as flow charts, conceptual diagrams, or schematic illustrations).
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We encourage the use of supplementary materials such as applications, images, and

- Supplementary material should be accurate and relevant to the research.
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- Submit all supplementary material at the same time as your manuscript.
- Include a concise caption for each supplementary file, describing its content.
- After submission, supplementary files can only be added or replaced in the revision stage of the editorial process.

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This journal accepts video material and animation sequences to support and enhance your scientific research. We encourage you to include links to video or animation files within articles. Some guidelines:

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- Provide files in one of our recommended file formats. Files should be within our preferred maximum file size of 150 MB per file, 1 GB in total.
- Provide "stills" for each of your files. These will be used as standard icons to personalize the link to your video data. You can choose any frame from your video or animation or make a separate image.
- Provide descriptive text in your manuscript to refer to the video content. This text helps ensure accessibility for visually impaired readers, and is essential for journals publishing in print.

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Research data

We are committed to supporting the storage of, access to and discovery of research data, and our research data policy sets out the principles guiding how we work with the research community to support a more efficient and transparent research process.

Research data refers to the results of observations or experimentation that validate research findings, which may also include software, code, models, algorithms, protocols, methods and other useful materials related to the project.

Please read our guidelines on sharing research data for more information on depositing, sharing and using research data and other relevant research materials.

Research data deposit, citation and linking

For this journal, **Option C** instructions from our research data guidelines apply. This means that you are **required** to:

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- Cite and link to this dataset in your article.
- If this is not possible, make a statement explaining why research data cannot be shared.

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Which articles connect with your research?

Describe your research question and surface relevant articles from this journal and related peer-reviewed literature.

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You are encouraged to publish a description of your research data, methods, or protocols related to your regular article as a co-submission article in Data in Brief or MethodsX.

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Note that due to the automated process of submitting to *Data in Brief* or *MethodsX*, you will only be presented with an APC after acceptance. This will be a personalized charge based on your individual context (e.g., your country, institutional affiliation, and society membership for example).

Please visit the co-submission Researcher Support page for more information on how to submit a co-submission.

Article structure

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- Divide your manuscript into clearly defined and numbered sections. Number subsections 1.1 (then 1.1.1, 1.1.2, ...), then 1.2, etc.
- Use the numbering format when cross-referencing within your article. Do not just refer to "the text."
- You may give subsections a brief heading. Headings should appear on a separate line.
- Do not include the article abstract within section numbering.

Introduction

The introduction should clearly state the objectives of your work. We recommend that
avoid writing a detailed literature overview or summary of your results.

Methods

The methods section should provide sufficient details about your materials and meth
independent researcher. Some guidelines:

- If the method you used has already been published, provide a summary and refer
- If you are quoting directly from a previously published method, use quotation marks and cite the source.

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Describe your research question and surface relevant articles from this journal and related peer-reviewed literature.

- Describe any modifications that you have made to existing methods.

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Results should be clear, concise and reproducible. We advise you to read the sections in this guide on supplying tables, artwork, supplementary material, and sharing research data.

Discussion

The discussion section should explore the significance of your results but not repeat them. Avoid the use of extensive citations and discussion of published literature in the discussion section.

Conclusion

The conclusion section should present the main conclusions of your study. You may have a stand-alone conclusions section or include your conclusions in a subsection of your discussion or results and discussion section.

Acknowledgements

Include any individuals who provided you with help during your research, such as help with language, writing, or proofreading, in the acknowledgments section.

Acknowledgments should be placed in a separate section that appears directly before the reference list. Do not include acknowledgments on your title page, as a footnote to your title, or anywhere else in your article other than in the separate acknowledgments section.

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Corresponding authors are required to acknowledge co-author contributions using CRediT (Contributor Roles Taxonomy) roles:

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Describe your research question and surface relevant articles from this journal and related peer-reviewed literature.

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We ask you to use the following format for appendices:

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- Give separate numbering to formulae and equations within appendices using formats such as Eq. (A.1), Eq. (A.2), etc. and in subsequent appendices, Eq. (B.1), Eq. (B.2) etc. In a similar way, give separate numbering to tables and figures using formats such as Table A.1; Fig. A.1, etc.

References

References within text

Any references cited in your article should also be present in your reference list and vice versa. Some guidelines:

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- Make sure all data provided in your reference list are correct and complete.
- Every reference included must correspond to a real, existing source.
- Where available include Digital Object Identifiers (DOIs), as they provide a permanent link to the electronic article referenced and improve the discoverability of your research.
- We recommend not including unpublished results or personal communications in your reference list, though you may mention them in the text of your article. If included, they must follow the standard reference style of the journal, substituting publication date with “unpublished results” or “personal communication”.
- References cited as “in press” imply that the item has been accepted for publication.

Reference format

This journal does not set strict requirements on reference formatting at submission. Some guidelines:

- References can be in any style or format as long as the style is consistent.
- Author names, journal or book titles, chapter or article titles, year of publication, volume numbers, article numbers, and pagination must be included, where applicable.

Our journal reference style will be applied to your article after acceptance, at proof stage. If required, at this stage we will ask you to correct or supply any missing reference data.

Reference style

Indicate references by number(s) in square brackets in line in the text. You can refer to author names within your text, but you must always give the reference number, e.g., "as demonstrated [3,6]. Barnaby and Jones [8] obtained a different result ....".

Number the references in the reference list in the order in which they appear in the text.

Please note the shortened form for last page number. e.g., 51-9, and that for more than 6 authors the first 6 should be listed followed by 'et al.'.

For further details you are referred to Recommendations for the Conduct, Reporting, Editing and Publication of Scholarly Work in Medical Journals and more samples of formatted references.

Abbreviate journal names according to the List of Title Word Abbreviations (LTWA).

Examples:

Reference to a journal publication:

[1] Van der Geer J, Hanraads JAJ, Lupton RA. The art of writing a scientific article. J Sci
https://doi.org/10.1016/j.sc.2020.00372.

Reference to a journal publication with an article number:

[2] Van der Geer J, Hanraads JAJ, Lupton RA. The art of writing a scientific article. Heliy
https://doi.org/10.1016/j.heliyon.2022.e00205.

Which articles connect with your research?

Describe your research question and surface relevant articles from this journal and related peer-reviewed literature.

Reference to a book:
[3] Strunk Jr W, White EB. The elements of style. 4th ed. New York: Longman; 2000.

Reference to a chapter in a book:
[4] Mettam GR, Adams LB. How to prepare an electronic version of your article. In: Jones BS, Smith RZ, editors. Introduction to the electronic age, New York: E-Publishing Inc; 2023, p. 281-304.

Reference to a website:
[5] Cancer Research UK. Cancer statistics reports for the UK, http://www.cancerresearchuk.org/aboutcancer/statistics/cancerstatsreport/; 2023 [accessed 13 March 2023].

Reference to a dataset:
[6] Oguro M, Imahiro S, Saito S, Nakashizuka T. Mortality data for Japanese oak wilt disease and surrounding forest compositions [dataset]. Mendeley Data. 2015. https://doi.org/10.17632/xwj98nb39r.1.

Reference to software:
[7] Coon E, Berndt M, Jan A, Svyatsky D, Atchley A, Kikinzon E, et al. Advanced Terrestrial Simulator (ATS). Version 0.88 [software]. Zenodo; 2020 Mar 25. https://doi.org/10.5281/zenodo.3727209.

Web references
When listing web references, as a minimum you should provide the full URL and the date when the reference was last accessed. Additional information (e.g., DOI, author names, dates, or reference to a source publication) should be provided if known.

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Which articles connect with your research?

Describe your research question and surface relevant articles from this journal and related peer-reviewed literature.

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Submitting your manuscript

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- All files have been uploaded, including artwork, videos, and supplementary materials. Don't forget to include tables and footnotes in the files, and to submit captions.
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- All references in the article text are cited in the reference list, and vice versa.
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